

COURSE TITLE : MASS TRANSFER OPERATIONS 1
COURSE CODE : 5071
COURSE CATEGORY : A
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 52
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Diffusion and Absorption	13
2	Humidification	13
3	Drying Operations	13
4	Distillation	13
TOTAL		52

COURSE OUTCOMES :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	understand the basic principle of mass transfer operations
	2	Comprehend different type of diffusion processes
	3	Understand the mechanism of absorption Analyse the material balance of gas absorption
2	4	Comprehend different type of humidity.
	5	Analyse the problems using humidity chart
	6	Understand the working of cooling towers
3	7	Understand the mechanism of drying Appreciate the operations of different driers Analyse the simple problems of drying
	8	Comprehend the principles of different distillation Appreciate the working of different distillation units Solve the problems relating to distillation.

SPECIFIC OUTCOME

MODULE I DIFFUSION AND ABSORPTION

1.1.0 Know the diffusion.

- 1.1.1 Define diffusion.
- 1.1.2 Classification of diffusion.
- 1.1.3 Define eddy diffusion.
- 1.1.4 Define molecular diffusion.
- 1.1.5 Define thermal diffusion.
- 1.1.6 Explain the molar flux.
- 1.1.7 Explain fick 's rate equation.

1.2.0 Understand the Absorption operations.

- 1.2.1 Define absorption.
- 1.2.2 Define desorption.
- 1.2.3 Define Henry's law.
- 1.2.4 Comparison of gas absorption and distillation.
- 1.2.5 Explain the mechanism absorption operation.
- 1.2.6 Explain the condition of equilibrium between gas and liquid.
- 1.2.7 Selection criteria for solvent in gas absorption.
- 1.2.8 Explain material balance relating absorption.
- 1.2.9 Solve simple problems relating to material balance.

1.3.0 Understand the construction and working of Absorption operation equipments.

- 1.3.1 Explain constructional details and working of packed tower with neat sketch.
- 1.3.2 Describe the different types of packing materials with example.
- 1.3.3 List the important characteristics of packing materials.
- 1.3.4 Distinguish between regular and random packing.
- 1.3.5 Explain the flow patterns of fluid through packing.
- 1.3.6 Define the term channelling.
- 1.3.7 Describe the various types of liquid distributors.

1.4.0 Understand the mechanism of adsorption in exchange process.

- 1.4.1 Differentiate between physical and chemical adsorption.
- 1.4.2 List the various adsorbents used and their applications.
- 1.4.3 Explain the manufacture of various adsorbents.
- 1.4.4 Explain the theory of adsorption.

MODULE II HUMIDIFICATION

2.1.0 Understand the Principles of Humidification

- 2.1.1 Define the terms – humidity, relative humidity percentage humidity, Molar humidity, humid volume & humid heat.
- 2.1.2 Identify the above terms in the form of a mathematical equation.
- 2.1.3 Solve problems based on the above equations.
- 2.1.4 Explain the humidity chart for air-water vapour system.
- 2.1.5 Calculate humidity, dew point, percentage humidity etc., from humidity chart.
- 2.1.6 Solve simple problems using humidity chart.
- 2.1.7 Explain the terms adiabatic saturation temp and wet bulb temp.
- 2.1.8 Derive the equation for adiabatic saturation temperature and wet bulb temperature.
- 2.1.9 Explain the factors influencing the adiabatic saturation and wet bulb temperature.

2.2.0 Understand the construction and working of water cooling towers.

- 2.2.1 Explain the constructional details of atmospheric type cooling.
- 2.2.2 Explain the constructional details of chimney type cooling tower.
- 2.2.3 Explain the constructional details of Forced draft cooling towers.

- 2.2.4 Explain the methods of humidification and dehumidification.
- 2.2.5 Describe the working of Humidifiers.
- 2.2.6 Compare the humidification and dehumidification operations.

MODULE III DRYING OPERATIONS

3.1.0 Understand the mechanism of drying operations.

- 3.1.1 List the industrial applications of drying.
- 3.1.2 Explain the classification of drying equipments.
- 3.1.3 Explain the constructional details and working of Tunnel dryer Rotary dryer, spray dryer Drum dryer Fluidised bed dryer and Tray dryer.
- 3.1.4 Explain the control measure adopted for the control of drying operation in the case of dryers.
- 3.1.5 List the field of application of various dryers.
- 3.1.6 Compare the performance of various dryers.
- 3.1.7 Explain the theory of drying.
- 3.1.8 Distinguish between wet basis and dry basis of expression of moisture content of solids.
- 3.1.9 Define the terms equilibrium moisture content, free moisture content and critical Moisture content.
- 3.1.10 Distinguish between bound water and unbound water.
- 3.1.11 Explain the mechanism of batch drying.
- 3.1.12 Explain constant rate period and falling rate period.
- 3.1.13 Derive an equation for calculating the time for drying (constant rate period and falling rate period)
- 3.1.14 Simple problems using the above derived equation.
- 3.1.15 Explain the factors influencing the rate of drying.

MODULE- IV

4.1.0 Understand the Basic Principles of Distillation.

- 4.1.1 List the applications of distillation operation.
- 4.1.2 Define the terms less volatile, more volatile, low boiling and high boiling.
- 4.1.3 Define Raoult's law.
- 4.1.4 Define and distinguish ideal and non ideal solutions.
- 4.1.5 Estimate compositions in terms of mole fractions.
- 4.1.6 Construct the vapour – liquid equilibrium diagram for a binary mixture.
- 4.1.7 Determine vapour liquid equilibrium data applying Raoult's law.
- 4.1.8 Derive an equation for relative volatility.
- 4.1.9 Solve the problems using the above derived equation.
- 4.1.10 Explain maximum and minimum boiling azeotrope with suitable examples.
- 4.1.11 List the various methods of distillation.
- 4.1.12 Explain equilibrium distillation with neat sketch.
- 4.1.13 Explain simple distillation.
- 4.1.14 Derive and verify Rayleigh's equation for simple distillation.
- 4.1.15 Solves the problem based on Rayleigh's equation.
- 4.1.16 Explain steam distillation.
- 4.1.17 List the advantages and applications of steam distillation.

CONTENT DETAILS

MODULE I

Diffusion

Molecular diffusion – molar flux – Fick's rate equation.

Absorption

Mechanism of Absorption – conditions of equilibrium between gas and liquid – Henry's law – factors controlling rate of absorption – simple problems – equipments for absorption operations – packed tower – packing materials – characteristics of packing – liquid distributors – channelling.

Adsorption

Physical and chemical adsorption – various types of adsorbents – applications – manufacture of adsorbents; ion exchange – principle — molecular sieves

MODULE II

HUMIDITY

General mechanism of diffusion processes – Definitions and mathematical expressions for molal humidity – Absolute humidity – Relative humidity – percentage humidity – Humid volume – Humid heat – Enthalpy and dew point – simple problems. Humidity chart – its importance and application in solving problems – simple problems

Adiabatic saturation temperature – wet bulb temp cooling towers – atmospheric – natural and forced draft, cross flow type – humidification and dehumidification – air conditioning

MODULE III

DRYING

Purpose and industrial applications – drying equipment – classifications – tray dryer – tunnel dryer – rotary dryer – turbo dryer – spray dryer – drum dryer – cylinder dryer – fluid bed dryer – dry basis and wet basis of expression of moisture content – equilibrium moisture content – free moisture content – bound and unbound water – mechanism of batch drying of solids – constant rate and falling rate period. Derivations of equations for time of drying in constant rate and falling rate period – simple problems.

MODULE IV

DISTILLATION

Distillation as an inter phase mass transfer – industrial application – definition of terms– less volatile, more volatile, low boiling, high boiling – vapour – liquid equilibrium diagrams and their importance. Ideal and non-ideal solutions – Raoult's law – calculation of X-Y data using Raoult's law. Azeotropes – maximum and minimum boiling – volatility and relative volatility – calculation of relative volatility of a binary mixture. Types of distillation

- **equilibrium – simple distillation- steam distillation.**

REFERENCE

- | | | |
|------------------|---|--|
| McCabe and Smith | - | Unit Operations |
| Treybal.R.E | - | Mass Transfer Operations |
| P.Chathopadhy | - | Unit Operations of Chemical Engg. Vol. – I |
| Surya Narayana | - | Mass Transfer Operations |
| K A Gavhane | - | Unit Operations. II |